

real-world Python Input/Output questions based on Arithmetic, Membership, Identity, Logical, Relational, and Bitwise operators

1. Arithmetic Operators (Real World)

1. Write a Python program to calculate total salary using basic salary and bonus.
Formula: $\text{Total Salary} = \text{Basic} + \text{Bonus}$
2. Write a Python program to calculate electricity bill using units and cost per unit.
Formula: $\text{Bill} = \text{Units} \times \text{Cost}$
3. Write a Python program to calculate area of a rectangle (length $\times$ width).
4. Write a Python program to calculate simple interest.
Formula: $\text{SI} = (\text{P} \times \text{R} \times \text{T}) / 100$
5. Write a Python program to calculate shopping bill after discount.
Formula: $\text{Final Price} = \text{Price} - \text{Discount}$

2. Relational Operators (Real World)

6. Write a Python program to check if a student passed or failed (marks ≥ 40).
7. Write a Python program to check if age is eligible for voting (age ≥ 18).
8. Write a Python program to check if a number is greater than 100.
9. Write a Python program to compare two product prices and find cheaper one.
10. Write a Python program to check if entered password length is valid (≥ 8).

3. Logical Operators (Real World)

11. Write a Python program to check login success (username AND password correct).
12. Write a Python program to check scholarship eligibility (marks > 80 AND attendance > 75).
13. Write a Python program to check discount eligibility (age > 60 OR student = True).
14. Write a Python program to check exam eligibility (attendance ≥ 75 AND fees paid).
15. Write a Python program to check loan approval (income $>$ limit AND no loan default).

4. Membership Operators (Real World)

16. Write a Python program to check if a product is available in stock list.
17. Write a Python program to check if a student is in topper list.
18. Write a Python program to check if entered city is in allowed delivery list.
19. Write a Python program to check if username exists in database list.
20. Write a Python program to check if a subject is in student's course list.

5. Identity Operators (Real World)

21. Write a Python program to check if two variables refer to the same object.
22. Write a Python program to check if two lists are identical objects or not.
23. Write a Python program to check if a variable is None.

24. Write a Python program to check if two employee records point to same memory.

25. Write a Python program to check if two strings are the same object.

6. Bitwise Operators (Real World)

26. Write a Python program to perform permission check using bitwise AND.

27. Write a Python program to toggle system settings using XOR.

28. Write a Python program to double a number using left shift.

29. Write a Python program to divide a number by 2 using right shift.

30. Write a Python program to check even or odd using bitwise AND.

7. Mixed Real-World Problems

31. Write a Python program to calculate net salary after tax and bonus.

32. Write a Python program to check if user is eligible for discount using age and membership status.

33. Write a Python program to validate login using username, password, and role.

34. Write a Python program to check if a student is eligible for exam using attendance and fees status.

35. Write a Python program to find best deal between two products using price comparison.

8. ML / Data Science Related Logic

36. Write a Python program to check if dataset column exists in feature list (membership).

37. Write a Python program to check if two model outputs are identical (identity).

38. Write a Python program to compare prediction accuracy of two models (relational).

39. Write a Python program to check feature selection using AND condition.

40. Write a Python program to simulate binary feature masking using bitwise operators.

9. BITWISE QUESTIONS:

Core Bit Manipulation (Must Do First)

1. Swap bits in a given number
2. Smallest of three integers without comparison operators
3. Compute absolute value without branching
4. Compute minimum or maximum without branching
5. Find the two non-repeating elements in an array
6. Reverse bits of a number
7. Smallest power of 2 greater than or equal to n
8. Find parity of an unsigned integer
9. Swap all odd and even bits
10. Check if binary representation is palindrome

11. Generate n-bit Gray Codes
12. Check if a number is sparse
13. Find nth Magic Number
14. Swap two bits in a given integer
15. Calculate square without using $*$, $/$, $\text{pow}()$
16. Copy set bits in a range
17. Gray to Binary and Binary to Gray conversion
18. Find Next Sparse Number
19. Sum of bit differences among all pairs
20. Bitwise AND of a range
21. Length of longest consecutive 1s in binary representation
22. Print first n numbers with exactly two set bits
23. Count trailing zero bits using lookup table
24. Multiplication using shift operator
25. Determine if a string has all unique characters
26. Count numbers whose sum with x equals XOR with x
27. Maximum XOR value of a pair
28. Next greater integer with one more set bit
29. Check if two numbers are bit rotations
30. Previous smaller integer with one less set bit

Very Important Array + XOR Problems

31. Unique element where all occur twice except one
32. Find unique pair in array with pairs of numbers
33. Find two missing numbers using XOR
34. Maximum subset XOR of a set
35. Count pairs with odd XOR
36. XOR of all subarray XORs
37. Queries on XOR of XORs of subarrays
38. Sum of Bitwise AND of all pairs
39. Count all pairs differing in K bits
40. Number of unique triplets whose XOR is zero

Frequently Asked Mathematical Bit Tricks

41. Divide two integers without using *, /, %
42. Increment a number using bits
43. Quotient and remainder dividing by 2^k
44. Highest power of $2 \leq n$
45. Check if number is power of 8
46. Check divisibility by 17 using bitwise operators
47. Check binary string multiple of 3 using DFA
48. Find two numbers from sum and XOR
49. Minimum bitwise operations to convert a into b
50. Minimum flips to maximize number with k set bits

Important String + Bitmask Problems

51. Efficiently check duplicates in string without extra DS
52. Print all subsequences of a string (Iterative Bitmask)
53. Case conversion using bitwise operators
54. Efficiently find first repeated character in one traversal
55. Subset sum queries using bitset

Advanced / High-Level Interview Problems

56. Fast inverse square root
57. Josephus Problem Using Bit Magic
58. Bitwise Sieve
59. Reverse bits using lookup table in $O(1)$
60. Space optimization using bit manipulations
61. Cyclic Redundancy Check (CRC) and Modulo-2 Division
62. Variation in Nim Game
63. Maximum AND value of a pair in array
64. Print pair with maximum AND value
65. Find profession in a special family

Most Repeated Interview Favorites

These are asked again and again in companies:

- Reverse bits of a number
- Count set bits
- Power of 2 checks

- XOR unique element problems
 - Swap odd/even bits
 - Gray code generation
 - Divide without operators
 - Maximum XOR pair
 - Bitwise AND of range
 - Josephus problem
 - Maximum subset XOR
 - Subsequence generation using bitmask
 - Fast inverse square root
 - Sparse number problems
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Best Order to Prepare

Beginner

- Power of 2
- Count set bits
- XOR basics
- Swap bits
- Reverse bits
- Odd/even bit swap

Intermediate

- Unique element using XOR
- Maximum XOR pair
- Gray codes
- Sparse numbers
- Bitwise AND range
- Subsequence generation

Advanced

- Maximum subset XOR
- Fast inverse square root
- CRC
- Nim Game

- Bitset optimization
- DFA divisibility problems